
TN 19.1.0.1: UserSet - A short Introduction

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1 Context and Challenge: Storing gen4 Device Configurations

Heliotis Gen4 devices offer highly customizable configurations through device features. In some cases, one may not want to specify the entire configuration for an inspection task in the control application.

Additionally, some configurations are explored in a lab environment before being applied in production. The standard approach is to implement the new configuration directly in the user application. However, in certain situations, modifying the user application is not feasible/permitted.

In these cases, it becomes necessary to store specific configurations directly on the device.

2 Solution: UserSet

To address these scenarios, Gen4 devices support the *UserSet* functionality, as defined by GenICam and described in the Standard Feature Naming Convention (SFNC) of GenICam¹. This technical note outlines various use cases and procedures for configuring Heliotis Gen4 devices through UserSets.

Heliotis devices offer two custom UserSets: *UserSet0* and *UserSet1*, which serve as independent storage locations for full device configurations. A third option, *Default*, is available through the *UserSetSelector* feature. The *Default* option is read-only and represents the factory default configuration.

The primary UserSet operations include:

¹ EMVA GenICam SFNC: https://www.emva.org/wp-content/uploads/GenICam_SFNC_v2_7.pdf

- *Store* – Saves the current configuration to the selected storage location on the device.
- *Load* – Loads a previously stored configuration from the selected storage location.

Additionally, users can define which UserSet (*UserSet0*, *UserSet1*, or *Default*) should automatically load upon device boot-up or reset.

2.1 Storing (or Creating) a UserSet

To *store* a UserSet, an application or script must connect to the Gen4 device and configure it appropriately. Matching feature values can be determined using a GenICam Viewer or the Heliotis-specific heliViewer. Further recommendations may be provided by Heliotis.

Important: A UserSet must be stored while the device is in configuration mode (i.e. not during an acquisition) and before the device connection is closed (which triggers a reset).

1. First, select the desired UserSet using `UserSetSelector`.
 2. Then, execute the corresponding save operation (e.g., `UserSetSave`).
-

Listing 2.1: Store a UserSet

```
UserSetSelector = "UserSet0"
UserSetSave.execute()
```

2.2 Loading a UserSet

To apply a previously stored UserSet, the application must execute the Load operation after establishing a device connection. This process loads the full camera configuration, overwriting any previously set feature values.

Listing 2.2: Load a UserSet

```
UserSetSelector = "UserSet0"
UserSetLoad.execute()
```

2.3 Default UserSet

Gen4 devices support configuring a *Default UserSet* that loads automatically upon boot-up, reset, or reconnection. This allows users to define default values for the device.

The feature name for configuring the *Default UserSet* is `UserSetDefault`. Unlike most other features, `UserSetDefault` is non-volatile, meaning its value remains unchanged even after device resets, reconnections, or firmware updates.

Available options for `UserSetDefault`:

- `UserSet0`
- `UserSet1`
- `Default` (factory default setting)

Listing 2.3: Configure the default UserSet

```
UserSetDefault = "UserSet0"
```

2.4 File Transfer

Users may need to transfer a stored *UserSet* from one device to another. Gen4 devices support *UserSet* transfers through the *FileAccess* interface, as defined by GenICam.

Most GenICam-compatible applications support file transfer. Additionally, Heliotis provides *updateTool*, which facilitates *UserSet* transfers to and from devices.

Within *updateTool*, users can access the *Manage Files* dialog to select a device and define the desired *UserSet* through *FileSelector*. The transfer process from and to the file specified in *Local File* can then be initiated using *Write to Device* or *Read from Device*.

See also (see [Check File Transfer](#))

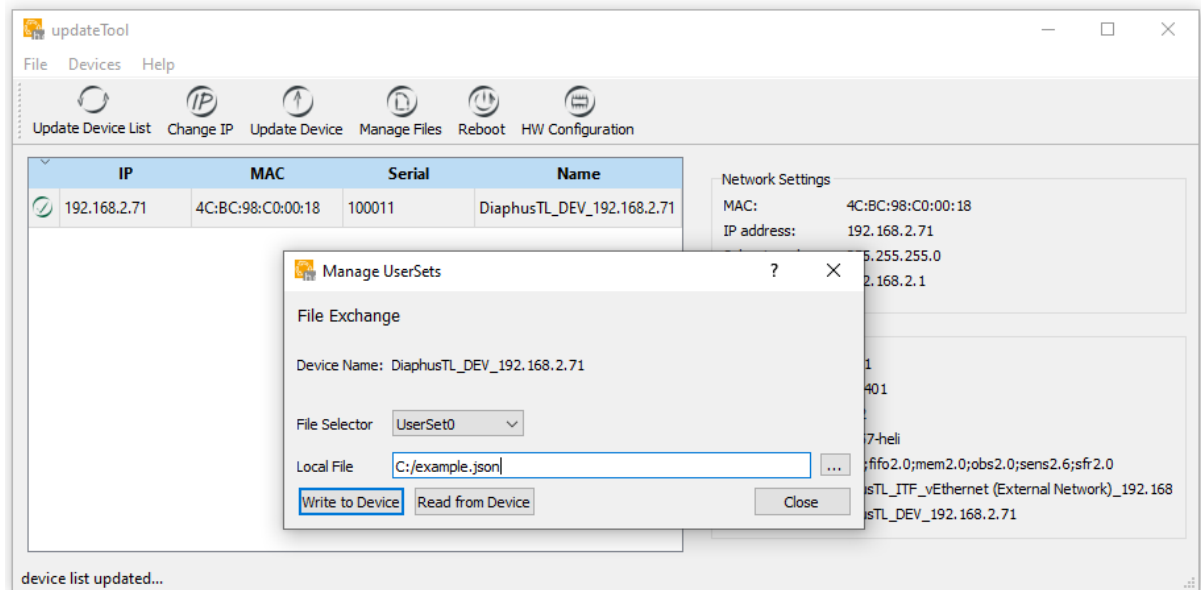


Fig. 2.1: File Transfer support in the updateTool

Limitations

1. The target device must have the exact same hardware configuration as the source device, since certain features depend on hardware compatibility.

- Example: External Xenax stages require IP address definitions that other stage types do not support.

2. The target device must have firmware that is the same or newer than the source device's firmware.

- While older UserSets can be migrated to newer firmware versions during file transfer, newer UserSets cannot be read by older firmware.

3. The File Transfer process does not provide an error notification when an unsupported UserSet is sent to a device.

- see [Check File Transfer](#)

2.5 Additional Features

AutomatedStageInit

When using heliInspect, the initial configuration after a power-up requires executing the `StageInit` command to initialize the stage. This procedure is not automated by default since additional configuration steps (e.g., defining an IP address) may be required.

With the *UserSet* mechanism, the full configuration - including IP address - becomes available at boot time, allowing automated stage initialization. Users can enable this feature by setting `AutomatedStageInit = true`.

Check File Transfer

The File Transfer implementation does not generate an error message if an unsupported *UserSet* is transferred to a device.

To verify whether the transfer was successful, users can access *log* information using the *File Manager* in *updateTool*.

1. Select Read from Device to retrieve the logs.
2. Store the logs as a .tar.gz file on the host computer.
3. Open the log file to check for error messages.

Listing 2.4: Example of an Error message when the *UserSet* transfer fails

```
[185582.447516]      Global 1956635728      ERROR   Failed to update UserSet0:␣  
↪ Could not convert file /data/heliService/userSets/userSet0.json
```

3 Document History

Version	Date	Changes	Responsible
19.1.0.0	22-11-2024	initial version	sm
19.1.0.1	12-06-2025	editing	bm